Program 3

3. Consider the Insurances database given below. The primary keys are underlined and the data types are specified.

PERSON (DRIVER-ID#: string, name: string, address: string)

CAR (Regno: string, model: string, year: int)

ACCIDENT (report-number: int, date: date, location: string)

OWNS (#driver-id: string, #Regno: string)

PARTICIPATED (#driver-id: string, #Regno: string, #report-number: int, Damage amount: int)

i. Create the above tables by property specifying the primary keys and the foreign keys.
ii. Enter at least five tables for each relation.
iii. Demonstrate how you

a. Update the damage amount for the car with a specific Regno in the accident with report number 12 to 25000.
b. Add a new accident to the database.

iv. Find the total number of people who owned cars that were involved in accidents in 2002.
v. Find the total number of accidents in which cars belonging to a specific model were involved.

i. Create the above tables by property specifying the primary keys and the foreign keys.

```
mysql> create database labprog3;
Query OK, 1 row affected (0.01 sec)

mysql> use labprog3;
Database changed

mysql> create table person(driver_id  varchar(20),
   ->  name  varchar(20),   address  varchar(20),  primary key(driver_id));
Query OK, 0 rows affected (0.09 sec)

mysql> create table car(regno  varchar(20),
   ->   model varchar(20),   year int,  primary key(regno));
Query OK, 0 rows affected (0.05 sec)

mysql> create table accident (report_no int,
   ->  accdate date,    location  varchar(20),   primary key(report_no));
Query OK, 0 rows affected (0.03 sec)

mysql>  create table owns(driver_id  varchar(20),   regno  varchar(20),
   ->  foreign key(driver_id) references person(driver_id),
   ->  foreign key(regno) references car(regno));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table participated (driver_id  varchar(20),
    ->   regno  varchar(20),    report_no int,   damage_amount  int,
    ->   foreign key(driver_id) references person(driver_id),
    ->   foreign key(regno) references car(regno),
    ->   foreign key(report_no) references accident (report_no));
Query OK, 0 rows affected (0.05 sec)
```

ii. Enter at least five tables for each relation.

```
mysql>  insert into person values('d001','kiran','hubli');
Query OK, 1 row affected (0.03 sec)

mysql>  insert into person values('d002','john','dharwad');
Query OK, 1 row affected (0.02 sec)

mysql>  insert into person values('d003','balaraj','belgaum');
Query OK, 1 row affected (0.03 sec)

mysql>  insert into person values('d004','siddu','chennai');
Query OK, 1 row affected (0.01 sec)

mysql>  insert into person values('d005','ravi','bangalore');
Query OK, 1 row affected (0.02 sec)

mysql> select * from person;
+----------+----------+---------------+
| driver_id | name     | address       |
+----------+----------+---------------+
| d001      | kiran    | hubli         |
| d002      | john     | dharwad       |
| d003      | balaraj  | belgaum       |
| d004      | siddu    | chennai       |
| d005      | ravi     | bangalore     |
+----------+----------+---------------+
5 rows in set (0.00 sec)

mysql> insert into car values('ka401','santro',2001);
Query OK, 1 row affected (0.03 sec)

mysql>  insert into car values('ka402','accent',2002);
Query OK, 1 row affected (0.00 sec)

mysql>  insert into car values('ka403','innova',2002);
Query OK, 1 row affected (0.00 sec)

mysql> insert into car values('ka404','accent',2003);
Query OK, 1 row affected (0.00 sec)

mysql>  insert into car values('ka405','santro',2005);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from car;

+-------+---------+--------+
| regno | model   | year   |
+-------+---------+--------+
| ka401 | santro | 2001   |
| ka402 | accent | 2002   |
| ka403 | innova | 2002  |
| ka404 | accent | 2003   |
| ka405 | santro | 2005   |
+-------+---------+--------+
5 rows in set (0.00 sec)

mysql> insert into accident values(20,'2002-01-12','kerala');
Query OK, 1 row affected (0.00 sec)

mysql>  insert into accident values(22,'2000-12-19','goa');
Query OK, 1 row affected (0.01 sec)

mysql>  insert into accident values(24,'2002-02-19','mysore');
Query OK, 1 row affected (0.02 sec)

mysql>  insert into accident values(12,'2005-07-20','londa');
Query OK, 1 row affected (0.00 sec)

mysql> insert into accident values(40,'2002-11-19','pune');
Query OK, 1 row affected (0.02 sec)

mysql>  select * from accident;
+-------------+---------------+---------------+
| report_no | accdate        | location       |
+-------------+---------------+---------------+
|       12       | 2005-07-20 | londa          |
|       20       | 2002-01-12 | kerala         |
|       22       | 2000-12-19 | goa            |
|       24       | 2002-02-19 | mysore        |
|       40       | 2002-11-19 | pune           |
+-------------+---------------+---------------+
5 rows in set (0.00 sec)

mysql> insert into owns values('d001','ka401');
Query OK, 1 row affected (0.05 sec)

mysql>  insert into owns values('d002','ka402');
Query OK, 1 row affected (0.01 sec)

mysql>  insert into owns values('d003','ka403');
Query OK, 1 row affected (0.02 sec)

mysql>  insert into owns values('d004','ka404');
Query OK, 1 row affected (0.00 sec)
mysql>  insert into owns values('d005','ka405');
Query OK, 1 row affected (0.00 sec)
```

```
mysql>  select * from  owns;
+-----------+-------+
| driver_id | regno |
+-----------+-------+
| d001      | ka401 |
| d002      | ka402 |
| d003      | ka403 |
| d004      | ka404 |
| d005      | ka405 |
+-----------+-------+
5 rows in set (0.00 sec)

mysql> insert into participated  values('d001','ka401',20,5000);
Query OK, 1 row affected (0.03 sec)

mysql>  insert into participated  values('d002','ka402',22,2000);
Query OK, 1 row affected (0.01 sec)

mysql>  insert into participated  values('d003','ka403',24,9000);
Query OK, 1 row affected (0.02 sec)

mysql>  insert into participated  values('d004','ka404',12,8000);
Query OK, 1 row affected (0.02 sec)

mysql>  insert into participated values('d005','ka405',40,800);
Query OK, 1 row affected (0.02 sec)

mysql> select * from participated;
+-----------+-------+-----------+----------------------+
| driver_id | regno | report_no | damage_amount |
+-----------+-------+-----------+----------------------+
| d001      | ka401 |     20    |        5000          |
| d002      | ka402 |     22    |        2000          |
| d003      | ka403 |     24    |        9000          |
| d004      | ka404 |     12    |        8000          |
| d005      | ka405 |     40    |         800          |
+-----------+-------+-----------+----------------------+
5 rows in set (0.00 sec)
```

iii. Demonstrate how you

a. Update the damage amount for the car with a specific Regno in the accident with report number 12 to 25000.

```
mysql> update participated set damage_amount=25000 where regno='ka404' and report_no=12;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

mysql> select * from participated;

driver_id	regno	report_no	damage_amount
d001	ka401	20	5000
d002	ka402	22	2000
d003	ka403	24	9000
d004	ka404	12	25000
d005	ka405	40	800

5 rows in set (0.00 sec)

b. Add a new accident to the database.

mysql> insert into accident values(50,'2006-01-13','kolhapur');
Query OK, 1 row affected (0.03 sec)

mysql> select * from accident;

report_no	accdate	location
12	2005-07-20	londa
20	2002-01-12	kerala
22	2000-12-19	goa
24	2002-02-19	mysore
40	2002-11-19	pune
50	2006-01-13	kolhapur

6 rows in set (0.00 sec)

iv. Find the total number of people who owned cars that were involved in accidents in 2002.

```
mysql> select count(*) from person, accident, participated
   ->  where  person.driver_id=participated.driver_id and
   ->  accident.report_no=participated.report_no and
   ->  accident.accdate between '2002-01-01' and '2002-12-31';
```

count(*)
3

1 row in set (0.00 sec)

v. Find the total number of accidents in which cars belonging to a specific model were involved.

```
mysql> select count(*) from car, accident, participated
   ->  where  car.regno=participated.regno and
   ->  accident.report_no=participated.report_no and  car.model='accent';
```

count(*)
2

1 row in set (0.00 sec)